THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM ONE
031 PHYSICS

Time: 2:30 Hours **Monday, 31st August 2026 a.m.**

Instructions

1. This paper consists of sections **A, B** and **C** with a total of **ten (10)** questions.
2. Answer **all** questions in the spaces provided.
3. Section **A** and **C** carry **fifteen (15)** marks each, Section B carries **seventy (70)** marks.
4. All writing must be in blue or black ink **except** drawings which must be in pencil.
5. All communication devices, calculators and any unauthorized materials are **not** allowed in the Assessment room.
6. Where necessary the following constants may be used:

   (i) Acceleration due to gravity, g = 10 N/kg
   (ii) Density of water = 1g /cm^3 or 1000kg/m^3

FOR ASSESSORS' USE ONLY		
QUESTION NUMBER	**SCORE**	**ASSESSOR'S INITIALS**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

SECTION A (15 MARKS)

1. For each of the items (i)-(x), choose the correct answer from the given alternatives and write its letter in the box provided.

(i) How does studying physics contribute to our understanding of the natural world and the universe?

A. Physics is not related to the study of the natural world and the universe
B. Physics helps us understand the behavior of matter and energy, from subatomic particles to celestial bodies
C. Physics only deals with artificial and human-made systems.
D. Physics is not essential for understanding the natural world and the universe.

(ii) When a large body of experimental evidence supports the hypothesis or does not supports the hypothesis, what may the hypothesis eventually be considered?

A. An observation
B. An insight
C. A conclusion
D. A law

(iii) Which of the following best describes a variable?

A. A trend that shows an exponential relationship.
B. Something whose value can change over multiple measurements
C. A measure of how much a plot line changes along the y-axis.
D. Something that remains constant over multiple measurements

(iv) Physics is the fundamental branch of science because the knowledge obtained in studying physics is strongly connected to other branches of science. Which among the following statements describes how physics is connected to other discipline like Chemistry and Biology?

A. Physics studies the properties of matter and energy, which are fundamental to all sciences.
B. Physics focuses solely on celestial bodies and space exploration
C. Physics deals with the human mind and cognitive processes
D. Physics has many applications in medical field.

(v) Physics is the study of matter, energy and the interaction existing between them. What is the significance of the interaction of matter and energy in Physics?

A. It helps us to understand the human body and its functions
B. It allows us to explore different planets and their atmospheres
C. It helps explain the behavior of objects and phenomena in the universe
D. It is not relevant to the study of Physics

(vi) To measure the length using a metre rule, wrong position of the eye leads to

A. Parallax error
B. Eye error
C. Zero error
D. Metre error

(vii) Seypalm was trying to measure the internal diameter of 100 ml beaker. Which of the

following instruments would you advise him to use for measuring the diameter?

A. Metre rule
B. Vernier caliper
C. Measuring tape
D. External caliper

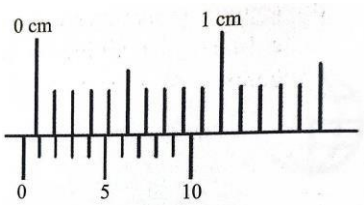
(viii) The mass of an object is reduced if its

A. State is changed
B. Amount of matter is reduced
C. Surface is reduced
D. Volume is reduced

(ix) A cuboid with length 3 cm, width 4 cm and height 10 cm is made from wood. The actual volume of the cuboid is

A. 120 cm^3
B. 240 ml
C. 120 m^3
D. 240 cm^3

(x) What is the zero-error shown in the figure below?



A. 0.3 mm
B. 0.7 mm
C. -0.3 mm
D. -0.7 mm

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)

2. Match the item in **List A** with the correct response **in List B** by writing the letter of the correct response below the corresponding item number provided in the table below.

LIST A	LIST B
(i) Stretching (ii) Attraction (iii) Friction (iv) Viscosity (v) Tension	A. Effect of magnet on iron material B. Force in string C. Extension of rigid material such as spring D. Two rough surfaces in contact E. Resistance to motion in fluid F. Week force G. The force that include both electric and magnetic force H. Exists in fluid and solid provided temperature remains constant I. It is an ideal force

(i)	(ii)	(iii)	(iv)	(v)

SECTION B (MARKS 70)

3. (a) Define the term ‘’Problem identification’’

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(b) When a ball is released from the top of a ramp, it rolls down to the bottom. After several trials, you notice that the ball always stops at almost the same point. How can you identify

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4. (a) Explain the meaning of the term ‘’hypothesis’’

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(b) You notice that when you rub a balloon against your hair, it becomes negatively charged, and you can make small pieces of paper stick to it. What testable hypothesis can you formulate to explain this phenomenon?

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5. (a) State **three effects of a force** on an object. **(3 marks)**

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(b) A spring has an original length of **20 cm**. When a force is applied, its length becomes **26 cm**.

(i) Calculate the **extension** of the spring. **(2 marks)**

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(ii) If the force is increased and the spring returns to its original length after the force is removed, what property of matter is demonstrated? **(1 mark)**

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(iii) State **two factors** that can affect the extension of an elastic material. **(2 marks)**

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(c) Explain why a stretched spring returns to its original shape when the stretching force is removed, provided its elastic limit has not been exceeded. **(2 marks)**

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6. (a) State the two factors that affect pressure exerted by a solid object on a flat surface.

(i) ……

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(ii) ……

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(b) A car accelerates uniformly from rest to a velocity of 20 m/s in 5 seconds. Calculate:

(i) The acceleration of the car.

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(ii) The total distance covered during this time.

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7. (a) Explain how the components of the screw gauge, include the sleeve and thimble readings, as well as the least count, are utilized to determine the precise diameter of the rod. What factors should be considered to ensure the accuracy of this of this measurement?

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(b) In the process of measuring the diameter of a rod using a micrometer screw gauge with a least count of 0.01 mm, the sleeve shows a reading of 1.6 cm, and the thimble indicates 48 divisions. Calculate the accurate diameter of the rod.

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8. (a) Explain the meaning of the word force?

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(b) List four naturally occurring forces

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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9. Explain four differences between mass and weight.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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SECTION C (15 MARKS)

10. (a) In each statement, suggest the type of force that comes into play:

(i) Two bar magnets push each other away.

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(ii) A book sliding across a table slowly comes to a stop.

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(iii) A dropped ball falls towards the ground.

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(b) Briefly explain four branches of physics.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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(c) A metal block of volume **200 cm³** is completely immersed in water. The density of water is **1 g/cm³**. If the block experiences an upthrust of **2 N**, state whether the block will **sink or float** when released. Give a reason for your answer. Take **g = 10 N/kg**.